Raj

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Profile

Recent MSc Data Science graduate with hands-on experience in Python, SQL, machine learning, data analysis, and LLM-based analytics systems. Built and evaluated machine learning models across predictive modelling and NLP projects, complemented by prior IT experience as a Scrum Master focused on stakeholder management, business problem-solving, cross-functional collaboration, and data-informed decision-making.

Technical Skills

Programming: Python, SQL, pandas, NumPy, PostgreSQL
Machine Learning: statistics, data analysis, data cleaning, EDA, feature engineering, ML fundamentals, scikit-learn, model evaluation, NLP, prompt engineering
Tools: Git, Kaggle, Azure DevOps, Visual Studio Code, Streamlit

Education

MSc Data Science, Aston University, Birmingham, UK

B.Tech in Information Technology, Anna University, India

Projects

Industry Dissertation: LLM-based Text-to-SQL Assistant

- Built a GenAI-powered Text-to-SQL chatbot system to automate SQL generation from natural-language questions on operational sensor data, automating manual query writing for operational analysis.
- Designed and evaluated context grounding and prompt engineering strategies to improve LLM-generated SQL reliability while managing accuracy, latency, and cost trade-offs under practical deployment constraints.

Stroke Risk Prediction System

- Built an end-to-end stroke risk prediction pipeline in Python using scikit-learn and imbalanced-learn, covering preprocessing, feature engineering, model development, comparative evaluation, and model selection on imbalanced clinical data; chose Balanced Random Forest for screening-focused performance in a high-cost false-negative healthcare use case.
- Deployed the model through a FastAPI API and Streamlit dashboard with Pydantic validation, Docker containerization, pytest-based testing, and feature-level explanation support.

Professional Experience

HealthAsyst, Scrum Master | Aug 2022 - Present

- Improved workflow efficiency across 6+ cross-functional teams with members in the US, UK, and India, reducing feature cycle time from 5-6 sprints to under 2 sprints through dependency tracking, business-value prioritisation, and data-informed planning.
- Used Azure DevOps and feature-level cycle time data to reduce feature turnaround from about 6 sprints to 1-2 sprints and improve focus on higher business-value outcomes.
- Improved visibility of dependencies, blockers, and delivery risks, reducing unplanned production issues to about one per sprint and improving release predictability.
- Built a custom AI agent, prompt repository, and supporting documentation to improve knowledge access, reduce cross-team coordination overhead, and enable more effective use of AI tools.

Certifications: Project Management Professional (PMP) | Professional Scrum Master I (PSM I) | Professional Scrum Master II (PSM II)